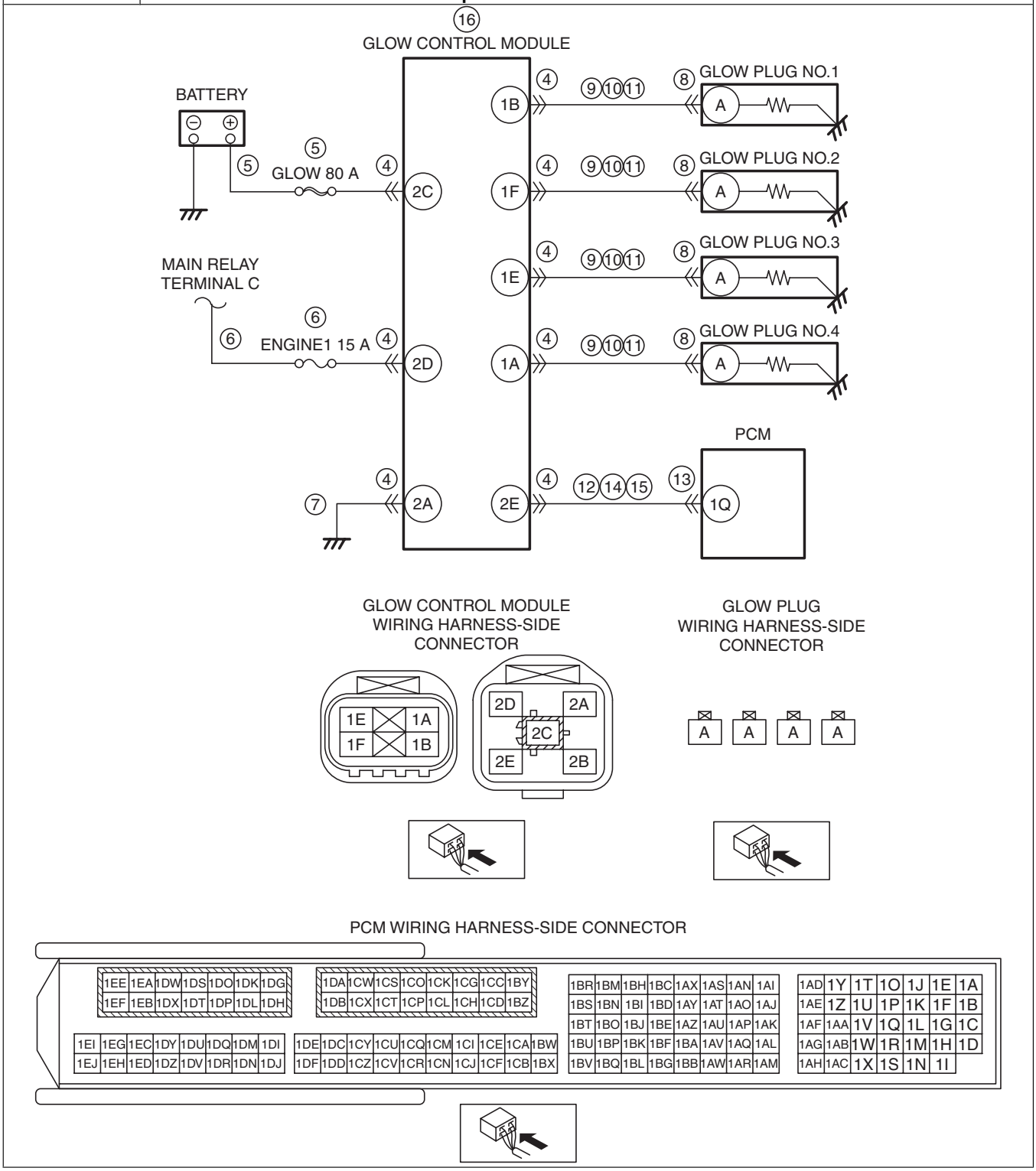
DTC P0670:00 [SKYACTIV-D 1.5]

id0102q2067000

DTC P0670:00	Glow control module control circuit problem
DETECTION CONDITION	• When the following conditions are met, a condition continues for **5 s** in which the power supply voltage of the glow control module is **less than 5.8 V** or **exceeds 16.5 V**, or there is an internal circuit malfunction, or the temperature of the internal circuit exceeds the specification. **MONITORING CONDITIONS** — 0.5 s have elapsed after ignition was switched ON. (engine off or on) — During glow control — Battery voltage: **8—20 V** — Glow plug No.1 contol circuit voltage: **3—11.3 V** — The following DTC is not detected: • LIN communication system: U0106:00 **Diagnostic support note** • This is an intermittent monitor (CCM). • The check engine light illuminates if the PCM detects the above malfunction condition during the first drive cycle. • FREEZE FRAME DATA/Snapshot data is available. • DTC is stored in the PCM memory.
FAIL-SAFE FUNCTION	• Not applicable
POSSIBLE CAUSE	• Glow control module connector or terminals malfunction • Short to ground or open circuit in glow control module power supply circuit — Short to ground in wiring harness between GLOW 80 A fuse and glow control module terminal 2C — GLOW 80 A fuse malfunction — Open circuit in wiring harness between battery positive terminal and glow control module terminal 2C • Short to ground or open circuit in glow control module power supply circuit — Short to ground in wiring harness between ENGINE1 15 A fuse and glow control module terminal 2D — ENGINE1 15 A fuse malfunction — Open circuit in wiring harness between main relay terminal D and glow control module terminal 2D • Open circuit in wiring harness between glow control module terminal 2A and body ground • Glow plug connector or terminals malfunction • Short to ground in wiring harness between the following terminals: — Glow control module terminal 1B—Glow plug No.1 terminal A — Glow control module terminal 1F—Glow plug No.2 terminal A — Glow control module terminal 1E—Glow plug No.3 terminal A — Glow control module terminal 1A—Glow plug No.4 terminal A • Short to power supply in wiring harness between the following terminals: — Glow control module terminal 1B—Glow plug No.1 terminal A — Glow control module terminal 1F—Glow plug No.2 terminal A — Glow control module terminal 1E—Glow plug No.3 terminal A — Glow control module terminal 1A—Glow plug No.4 terminal A • Glow plug circuits are shorted to each other • Short to ground in wiring harness between glow control module terminal 2E—PCM terminal 1Q • PCM connector or terminals malfunction • Short to power supply in wiring harness between glow control module terminal 2E—PCM terminal 1Q • Open circuit in wiring harness between glow control module terminal 2E—PCM terminal 1Q • Glow control module temperature is too high (**more than 123 °C {254 °F}**) • Glow control module malfunction • PCM malfunction

DTC P0670:00	Glow control module control circuit problem



Diagnostic Procedure

STEP	INSPECTION		ACTION
1	**RECORD VEHICLE STATUS WHEN DTC WAS DETECTED TO UTILIZE WITH REPEATABILITY VERIFICATION** **Note** • Recording can be facilitated using the screen capture of the PC function. • Record the freeze frame data/snap shot data.	—	Go to the next step.

STEP	INSPECTION		ACTION
2	**VERIFY RELATED SERVICE INFORMATION AVAILABILITY** • Verify related Service Information availability. • Is any related Service Information available?	Yes	Perform repair or diagnosis according to the available Service Information. • If the vehicle is not repaired, go to the next step.
		No	Go to the next step.
3	**VERIFY RELATED PENDING CODE AND/OR DTC** • Switch the ignition off, then ON (engine off). • Perform the Pending Trouble Code Access Procedure and DTC Reading Procedure. (See ON-BOARD DIAGNOSTIC TEST [SKYACTIV-D 1.5].) • Are any other PENDING CODEs and/or DTCs present?	Yes	Go to the applicable PENDING CODE or DTC inspection. (See DTC TABLE [SKYACTIV-D 1.5].)
		No	Go to the next step.
4	**INSPECT GLOW CONTROL MODULE CONNECTOR CONDITION** • Switch the ignition off. • Disconnect the glow control module connector. • Inspect for poor connection (such as damaged/ pulled-out pins, corrosion). • Is there any malfunction?	Yes	Repair or replace the connector and/or terminals, then go to Step 17.
		No	Go to the next step.
5	**INSPECT GLOW CONTROL MODULE POWER SUPPLY CIRCUIT FOR SHORT TO GROUND OR OPEN CIRCUIT** • Verify that the glow control module connector is disconnected. • Measure the voltage at the glow control module terminal 2C (wiring harness-side). • Is the voltage **B+**?	Yes	Go to the next step.
		No	Inspect the GLOW 80 A fuse. • If the fuse is blown: — Refer to the wiring diagram and verify whether or not there is a common connector between GLOW 80 A fuse and glow control module terminal 2C. **If there is a common connector:** • Determine the malfunctioning part by inspecting the common connector and the terminal for corrosion, damage, or pin disconnection, and the common wiring harness for a short to ground. • Repair or replace the malfunctioning part. **If there is no common connector:** • Repair or replace the wiring harness which has a short to ground. • Replace the fuse. • If the fuse is damaged: — Replace the fuse. • If the fuse is normal: — Refer to the wiring diagram and verify whether or not there is a common connector between battery positive terminal and glow control module terminal 2C. **If there is a common connector:** • Determine the malfunctioning part by inspecting the common connector and the terminal for corrosion, damage, or pin disconnection, and the common wiring harness for an open circuit. • Repair or replace the malfunctioning part. **If there is no common connector:** • Repair or replace the wiring harness which has an open circuit. Go to Step 17.

STEP	INSPECTION		ACTION
6	**INSPECT GLOW CONTROL MODULE POWER SUPPLY CIRCUIT FOR SHORT TO GROUND OR OPEN CIRCUIT** • Verify that the glow control module connector is disconnected. • Switch the ignition ON (engine off). • Measure the voltage at the glow control module terminal 2D (wiring harness-side). • Is the voltage **B+**?	Yes	Go to the next step.
		No	Inspect the ENGINE1 15 A fuse. • If the fuse is blown: — Refer to the wiring diagram and verify whether or not there is a common connector between ENGINE1 15 A fuse and glow control module terminal 2D. **If there is a common connector:** • Determine the malfunctioning part by inspecting the common connector and the terminal for corrosion, damage, or pin disconnection, and the common wiring harness for a short to ground. • Repair or replace the malfunctioning part. **If there is no common connector:** • Repair or replace the wiring harness which has a short to ground. • Replace the fuse. • If the fuse is damaged: — Replace the fuse. • If the fuse is normal: — Refer to the wiring diagram and verify whether or not there is a common connector between main relay terminal D and glow control module terminal 2D. **If there is a common connector:** • Determine the malfunctioning part by inspecting the common connector and the terminal for corrosion, damage, or pin disconnection, and the common wiring harness for an open circuit. • Repair or replace the malfunctioning part. **If there is no common connector:** • Repair or replace the wiring harness which has an open circuit. Go to Step 17.
7	**INSPECT GLOW CONTROL MODULE GROUND CIRCUIT FOR OPEN CIRCUIT** • Verify that the glow control module connector is disconnected. • Switch the ignition off. • Inspect for continuity between glow control module terminal 2A (wiring harness-side) and body ground. • Is there continuity?	Yes	Go to the next step.
		No	Refer to the wiring diagram and verify whether or not there is a common connector between glow control module terminal 2A and body ground. **If there is a common connector:** • Determine the malfunctioning part by inspecting the common connector and the terminal for corrosion, damage, or pin disconnection, and the common wiring harness for an open circuit. • Repair or replace the malfunctioning part. **If there is no common connector:** • Repair or replace the wiring harness which has an open circuit. Go to Step 17.
8	**INSPECT GLOW PLUG CONNECTOR CONDITION** • Disconnect the glow plug connector. • Inspect for poor connection (such as damaged/ pulled-out pins, corrosion). • Is there any malfunction?	Yes	Repair or replace the connector and/or terminals, then go to Step 17.
		No	Go to the next step.

STEP	INSPECTION		ACTION
9	**INSPECT GLOW PLUG CIRCUIT FOR SHORT TO GROUND** • Verify that the glow control module and glow plug connectors are disconnected. • Inspect for continuity between the following terminals (wiring harness-side) and body ground: — Glow plug No.1 terminal A — Glow plug No.2 terminal A — Glow plug No.3 terminal A — Glow plug No.4 terminal A • Is there continuity?	Yes	Refer to the wiring diagram and verify whether or not there is a common connector between the following terminals: • Glow control module terminal 1B—Glow plug No.1 terminal A • Glow control module terminal 1F—Glow plug No.2 terminal A • Glow control module terminal 1E—Glow plug No.3 terminal A • Glow control module terminal 1A—Glow plug No.4 terminal A **If there is a common connector:** • Determine the malfunctioning part by inspecting the common connector and the terminal for corrosion, damage, or pin disconnection, and the common wiring harness for a short to ground. • Repair or replace the malfunctioning part. **If there is no common connector:** • Repair or replace the wiring harness which has a short to ground. Go to Step 17.
		No	Go to the next step.
10	**INSPECT GLOW PLUG CIRCUIT FOR SHORT TO POWER SUPPLY** • Verify that the glow control module and glow plug connectors are disconnected. • Switch the ignition ON (engine off). • Measure the voltage at the following terminals (wiring harness-side): — Glow plug No.1 terminal A — Glow plug No.2 terminal A — Glow plug No.3 terminal A — Glow plug No.4 terminal A • Is the voltage **0 V**?	Yes	Go to the next step.
		No	Refer to the wiring diagram and verify whether or not there is a common connector between the following terminals: • Glow control module terminal 1B—Glow plug No.1 terminal A • Glow control module terminal 1F—Glow plug No.2 terminal A • Glow control module terminal 1E—Glow plug No.3 terminal A • Glow control module terminal 1A—Glow plug No.4 terminal A **If there is a common connector:** • Determine the malfunctioning part by inspecting the common connector and the terminal for corrosion, damage, or pin disconnection, and the common wiring harness for a short to power supply. • Repair or replace the malfunctioning part. **If there is no common connector:** • Repair or replace the wiring harness which has a short to power supply. Go to Step 17.
11	**INSPECT GLOW PLUG CIRCUITS FOR SHORT TO EACH OTHER** • Verify that the glow control module and glow plug connectors are disconnected. • Switch the ignition off. • Inspect for continuity between each glow plug terminals A (wiring harness-side). • Is there continuity?	Yes	Refer to the wiring diagram and verify whether or not there is a common connector between the following terminals: • Glow control module terminal 1B—Glow plug No.1 terminal A • Glow control module terminal 1F—Glow plug No.2 terminal A • Glow control module terminal 1E—Glow plug No.3 terminal A • Glow control module terminal 1A—Glow plug No.4 terminal A **If there is a common connector:** • Determine the malfunctioning part by inspecting the common connector and the terminal for corrosion, damage, or pin disconnection, and the common wiring harness for a short to each other. • Repair or replace the malfunctioning part. **If there is no common connector:** • Repair or replace the wiring harness which has a short to each other. Go to Step 17.
		No	Go to the next step.

STEP	INSPECTION		ACTION
12	**INSPECT GLOW CONTROL MODULE CIRCUIT FOR SHORT TO GROUND** • Verify that the glow control module and glow plug connectors are disconnected. • Inspect for continuity between glow control module terminal 2E (wiring harness-side) and body ground: • Is there continuity?	Yes	If the short to ground circuit could be detected in the wiring harness: • Refer to the wiring diagram and verify whether or not there is a common connector between glow control module terminal 2E and PCM terminal 1Q **If there is a common connector:** — Determine the malfunctioning part by inspecting the common connector and the terminal for corrosion, damage, or pin disconnection, and the common wiring harness for a short to ground. — Repair or replace the malfunctioning part. **If there is no common connector:** — Repair or replace the wiring harness which has a short to ground. If the short to ground circuit could not be detected in the wiring harness: • Replace the PCM (short to ground in the PCM internal circuit). (See PCM REMOVAL/INSTALLATION [SKYACTIV-D 1.5].) Go to Step 17.
		No	Go to the next step.
13	**INSPECT PCM CONNECTOR CONDITION** • Disconnect the PCM connector. • Inspect for poor connection (such as damaged/ pulled-out pins, corrosion). • Is there any malfunction?	Yes	Repair or replace the connector and/or terminals, then go to Step 17.
		No	Go to the next step.
14	**INSPECT GLOW CONTROL MODULE CIRCUIT FOR SHORT TO POWER SUPPLY** • Verify that the glow control module and glow plug and PCM connectors are disconnected. • Switch the ignition ON (engine off). • Measure the voltage at the glow control module terminal 2E (wiring harness-side): • Is the voltage **0 V**?	Yes	Go to the next step.
		No	Refer to the wiring diagram and verify whether or not there is a common connector between glow control module terminal 2E and PCM terminal 1Q **If there is a common connector:** • Determine the malfunctioning part by inspecting the common connector and the terminal for corrosion, damage, or pin disconnection, and the common wiring harness for a short to power supply. • Repair or replace the malfunctioning part. **If there is no common connector:** • Repair or replace the wiring harness which has a short to power supply. Go to Step 17.
15	**INSPECT GLOW CONTROL MODULE CIRCUIT FOR OPEN CIRCUIT** • Verify that the glow control module and glow plug and PCM connectors are disconnected. • Inspect for continuity between glow control module terminal 2E and PCM terminal 1Q (wiring harness-side): • Is there continuity?	Yes	Go to the next step.
		No	Refer to the wiring diagram and verify whether or not there is a common connector between glow control module terminal 2E and PCM terminal 1Q **If there is a common connector:** • Determine the malfunctioning part by inspecting the common connector and the terminal for corrosion, damage, or pin disconnection, and the common wiring harness for an open circuit. • Repair or replace the malfunctioning part. **If there is no common connector:** • Repair or replace the wiring harness which has an open circuit. Go to Step 17.
16	**INSPECT GLOW CONTROL MODULE** • Inspect the glow control module. (See GLOW PLUG CONTROL MODULE INSPECTION [SKYACTIV-D 1.5].) • Is there any malfunction?	Yes	Replace the glow control module, then go to the next step. (See GLOW PLUG CONTROL MODULE REMOVAL/ INSTALLATION [SKYACTIV-D 1.5].)
		No	Go to the next step.

STEP	INSPECTION		ACTION
17	**VERIFY DTC TROUBLESHOOTING COMPLETED** • Always reconnect all disconnected connectors. • Clear the DTC from the PCM memory using the M-MDS. (See AFTER REPAIR PROCEDURE [SKYACTIV-D 1.5].) • Perform the KOEO self test. (See KOEO/KOER SELF TEST [SKYACTIV-D 1.5].) • Is the same DTC present?	Yes	Repeat the inspection from Step 1. • If the malfunction recurs, replace the PCM. (See PCM REMOVAL/INSTALLATION [SKYACTIV-D 1.5].) Go to the next step.
		No	Go to the next step.
18	**VERIFY AFTER REPAIR PROCEDURE** • Perform the “AFTER REPAIR PROCEDURE”. (See AFTER REPAIR PROCEDURE [SKYACTIV-D 1.5].) • Are any DTCs present?	Yes	Go to the applicable DTC inspection. (See DTC TABLE [SKYACTIV-D 1.5].)
		No	DTC troubleshooting completed.